

Learning to learn

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NEWSPAPERS recently wrote about a crisis in engineering education.¹ It is reported that annually about 75,000 more engineering admission seats are going vacant because engineers are not getting jobs and that this is the result of poor quality of education. It should interest everyone that the nearly 72% of these colleges are private unaided that flourished over the last twenty-five years. But we do not hear many people talking about private sector being responsible for poor education. Anyway, the reason fewer people are going for engineering is because it does not end up in jobs and prosperity for a large number of graduates who are said to be unemployable. It is also true that with increasing automation, the industry needs either very highly skilled engineers or skilled shopfloor workers who need not be engineers. It would be interesting to see how many engineers are employed for their engineering degrees, if not knowledge.

Such exodus is not new. We saw it happen gradually out of municipal government schools in cities such as Mumbai. First, elite Marathi medium government-aided schools started losing students to English medium schools or started converting to English as medium of instruction. Next the municipal government schools that used to overflow with students once upon a time began emptying out starting at the southern tip of the city.

As we started measuring enrolment and reading/ math levels with the Annual Status of Education Report (ASER) survey, it became obvious that every year more and more children were moving to private schools. In 2016, as reported by DISE, the enrolment in private elementary schools is about 39%, which was just around 18% in 2007. It is not as though the private schools were there and children just switched. Just as the government was adding classrooms and appointing teachers, private operators sensed a new opportunity in the demand for education among the ordinary people as the economy grew rapidly. Private schools also started coming up in small towns and rural areas.

Today, more educated youth with high aspirations are finding it difficult to get jobs or get jobs that pay well. Many blame it on their lack of skills or lack of appropriate skills while others blame the economy. But what if masses of frustrated youth began to abandon the education system? ASER 2017, which surveyed young people between the ages of 14 and 18 found that although 85% were enrolled in schools or colleges, 40% worked for at least 15 days in the previous month. So, it is not clear what their enrolment means and how the institutions function. What if they were offered the alternative of not enrolling in institutions but instead in degree courses on-line, having access to notes and lectures (which do not change anyway), and directly appearing for examinations? It is not difficult to see that the colleges will empty out, although some may want to enrol so that they could hang out with friends. Or, perhaps there will be *dhabas* and cafes that will serve as places to hang out as they learn in groups? Then we will have a real crisis on our hands. But this is not possible. Or, is it?

It is commonly said that the economy is changing rapidly and the education system is not providing the kinds of skills to the children that are useful in the job market. But when did it ever? And how different are the foundational 21st century skills

from what worked well in the 20th century? Barring the specific need to understand and handle the new and rapidly changing digital technology, the skill of learning to learn, adaptability to a situation, ability to communicate, solving problems, and team work were always marketable skills.

The difference is that the school system never focused on these outcomes of education and doing well in academic subjects was considered the prime aim of good education. In the lead was (and is) a fearsome system of examinations that gives people nightmares decades after they have been liberated from that tyranny.

The phrase ‘learning outcomes’ has begun to go around a lot these days. I think different people understand it differently but it is easy to see that it is not the same as inputs, which we were used to for decades. As early as in 1990, the Jomtien Declaration cautioned us:

‘Whether or not expanded educational opportunities will translate into meaningful development – for an individual or for society – depends ultimately on whether people actually learn as a result of those opportunities, i.e., whether they incorporate useful knowledge, reasoning ability, skills, and values. The focus of basic education must, therefore, be on actual learning acquisition and outcome, rather than exclusively upon enrolment, continued participation in organized programmes and completion of certification requirements. Active and participatory approaches are particularly valuable in assuring learning acquisition and allowing learners to reach their fullest potential. It is, therefore, necessary to define acceptable levels of learning acquisition for educational programmes and to improve and apply systems of assessing learning achievement.’²

But, ten years later in 2000, the Dakar Framework for Action seems to have no memory of this article. The 6th and last goal of the framework was about improving quality of education, which was to be measured by the survival rate till grade 5. Little surprise then that the MDGs, also written up in 2000, set the target for ‘Goal 2 – Achieve Universal Primary Education’ as ‘Ensure that, by 2015, children everywhere, boys and girls alike, will be able to complete a full course of primary schooling’ and the indicators were net enrolment ratios and survival rates till grade 5.

The ’90s seem to have set off a number of national and international student assessments in the western world. But there was no quantitative information on the issue of quality in the developing world which was the target of the Education for All campaigns. Pratham’s Annual Status of Education Report provided, almost inadvertently, the first quantitative data point about ‘learning outcomes’ indicated by ability to read and to solve simple numeracy problems at a very low floor level in 2005-06. Close on its heels but completely independently, in 2006 the Early Grade Reading Assessment or EGRA was born in the West. The important commonality about the two, apart from testing at very low levels, is that the reading assessment is conducted orally one on one with the child and it is not a pen and paper test unlike the others.

Over the last ten to fifteen years a lot of quantitative data on basic learning outcomes of reading and numeracy have become available around the developing world through the application of ASER and EGRA. In India, the work done by Education Initiatives has contributed to this information and more recently the NCERT has conducted massive pen and paper assessments with results available at the district level. Although none of these surveys and studies are comparable with one another, everyone agrees that they all point to poor status of basic academic learning outcomes.

The world seems to have suddenly become aware that learning out-comes are poor all around. The phrase ‘schooling ain’t learning’ is now popular. Some decades ago it was said that years of schooling of its population is correlated with a nation’s economic growth. Now there is data to support the thought that it is not the years of schooling but the assessment scores or learning outcomes that are correlated with economic progress.

The World Development Report of 2018 has focused entirely on ‘learning’ and said we are in a ‘learning crisis’ situation. Its three point action formula is:

First, assess learning to make it a serious goal. Information itself creates incentives for reform, but many countries lack the right metrics to measure learning.

Second, act on evidence to make schools work for learning. Great schools build strong teacher-learner relationships in classrooms. As brain science has advanced and educators have innovated, the knowledge of how students learn most effectively has greatly expanded. But the way many countries, communities and schools approach education often differs greatly from the most promising, evidence-based approaches.

Third, align actors to make the entire system work for learning. Innovation in classrooms will not have much impact if technical and political barriers at the system level prevent a focus on learning at the school level. This is the case in many countries stuck in low-learning traps; extricating them requires focused attention on the deeper causes.

I liked the World Development Report 2018 not only because it favours Pratham’s kind of thinking, but because it is much broader and provides a good balanced picture of the situation and possible actions. It is available online for those wish to read it. All of us who are concerned about the status of education and learning outcomes are continually thinking of how to improve the system. It seems like this should be possible. But is it? Or is it too late?

This is the first time in the history of mankind that over 90% children in all countries are enrolled in formal schools with curricula, textbooks, teachers, classrooms, uniforms and examinations. This global rolling out of the mass education model, which was born in the late nineteenth century, has been happening over the last hundred, or at least seventy, years around the world. However, its efficiency and effectiveness was never questioned or measured seriously. It was assumed that if you open schools and provide all of the above, children will have the opportunity to get education, which was not incorrect. Clearly rise in literacy, availability of large numbers of educated people to perform many functions in growing populations and wealth were all outcomes of the expanding education system. There was reason for it to keep growing from the economic point of view and also from a social justice viewpoint.

The population of India was about 36 crore in 1951 and primary school enrolment was reported at 1.92 crore. Upper primary was 31 lakh. Today we have a population of about 130 crore and enrolment of 13.5 cr in primary and 6.5 cr in upper primary, making for an enrolment of around 96% children in this age group. In 2015-16, about 38% elementary schoolchildren went to private schools. In rural areas this proportion was around 30% and in urban around 57%.

This massive expansion of the school system in the far corners of our country may have had several objectives but as far as ordinary people were concerned it was all about ‘getting education’ to become a *padha likha insaan* and get a job, get out of poverty. The ‘padha likha insaan’ has some cultural socio-emotional attributes which should also count as outcomes of schooling but are not measured.

Ultimately, in the eyes of the *aam aadmi*, getting a good job is the main purpose of education. We have also been saying that if you get good education you will get a job. But is that true?

The table below shows how the number of students appearing for and passing std 10 and std 12 exams increased over a decade. Every year about 10 million students who clear their Higher Secondary School certificate examination enter undergraduate courses. Nearly 3.5 million go for BA, 1.5 million for BSc and another 1.3 million for BCom. These constitute about 60% of the students who come through the ‘quality check’ of the great education factory. Out of the remaining 4 million or so, 1.4 million seem to go to study various branches of technology and engineering. The remaining approximately 2.5 million new entrants go to some 50 odd courses including different branches of medicine.

Growing Number of Students and Improving Performance						
	Secondary Examinations			Higher Secondary Examinations		
Year	Appeared	Passed	Pass %	Appeared	Passed	Pass %
2005	1,34,86,740	86,09,855	64	78,28,610	53,86,899	72
2006	1,40,11,807	95,08,069	68	84,32,793	61,31,458	73
2007	1,47,12,429	1,03,44,687	70	88,68,155	65,66,178	74
2008	1,50,06,145	1,01,08,964	67	97,69,571	71,35,304	73
2009	1,70,89,658	1,18,19,689	69	1,05,51,871	80,44,636	76
2010	1,72,49,966	1,28,22,384	74	1,07,16,275	81,65,657	76
2011	1,80,24,161	1,34,89,768	75	1,15,67,858	87,11,254	75
2012	1,85,52,207	1,41,05,400	76	1,27,64,227	1,00,24,785	79
2013	2,03,15,906	1,57,97,305	78	1,40,59,950	1,10,34,079	78
2014	1,94,21,343	1,53,78,252	79	1,49,34,781	1,18,68,725	79
2015	1,91,16,429	1,50,89,278	79	1,43,23,497	1,13,41,644	79
Compiled from: http://mhrd.gov.in/statist						

In sum total, this is how the picture looks. Out of about 25 million children who enter std 1 every year, approximately 10 million or 40% will not go past std 10, one way or another. Of the 15 million who want to study more, half or 7.5 million will not graduate. Out of the 7.5 million who do graduate, how many are employable by the organized industry? It is said that 75% of fresh engineering graduates are not employable. So, we can easily take at least 50% as the not employable figure. That reduces the number of graduates employable in the organized sector down to about four million or less every year. This leads us to the next question – is the economy likely to generate four million new jobs every year that can absorb these graduates?

But this does not mean that the other 20-21 million remain unemployed. A large proportion may be underemployed. What is interesting is that government reported unemployment for those with school education is 3% while that for graduates is 10% and above graduation level is 18%.³ This is very similar to the reported unemployment profile of the Chinese youth.⁴ The numbers do not say if the

employment is appropriately matched with the pre-employment training or not. In all probability it is not. I think, in most cases people with general abilities are employed who then learn the specifics on the job.

But what kind of jobs are these? Are these the kind of jobs for which you need to spend 10 to 15 years in schools and colleges? Have we been wasting everybody's time?

In the middle of all political sloganeering these days, have you noticed that nobody is promising jobs? The euphoria over 'demographic dividend' is not noticeable anymore either. It is increasingly becoming apparent that new job creation in the old way, whether in the manufacturing sector or the financial services sector, will be very difficult.

In fact, chances are jobs will be lost and with those lost jobs will vanish the main reasons for why children, especially older children, should attend schools and colleges in this education system. A new one will have to be invented or perhaps it will just come together as the old schools and colleges, government or private, irrespective of medium of instruction, will start emptying out.

I know what you are thinking: This cannot be. Sometimes I think that too.

A friend directed me to a remarkable article by Kai-Fu Lee, an accomplished artificial intelligence innovator, computer scientist, who is now a venture capitalist. I reproduce some portions from the article that helps me make a point.

'We're all going to face a very challenging next fifteen or twenty years, when half of the jobs are going to be replaced by machines. Humans have never seen this scale of massive job decimation...

'The industrial revolution took a lot longer. And the industrial revolution created jobs while it replaced jobs. When it took a few artisans months to create a car, an automobile, an assembly line allowed that to happen in a fraction of the time by dividing the work into little chunks. Some jobs disappeared. Many jobs were created. Car prices came down. And then the job employment rate went up.

'Artificial intelligence is different because when we make a (machine) loan officer that decides whether to give someone a loan or not based on purely quantitative information, that loan officer will be better than 99 per cent of all loan officers out there. They would be replaced outright because it is a simple, single domain optimization problem – feed everything we know about a person in, and out comes the likelihood of repayment versus default. That rate is a quantitative computation based on a huge amount of data that no human can possibly match. The people in those jobs will be out of jobs and will have to do something else – same with security, with paralegal, with accounting, and even with reporters and translators...

'...As we think about all the benefits from AI, there are a number of issues one needs to be concerned about. One issue, as I talked about, was the job losses and how to deal with that. Another issue is the haves and have nots. The people who are inventing these AI algorithms, building AI companies, they will become the haves. The people whose jobs are replaced will be the have nots. And the gap between them, whether it's in wealth or power, will be dramatic, and will be perhaps the largest that mankind has ever experienced...'⁵

If what Kai-Fu Lee predicts is broadly true, the current school system will be wholly inadequate for the education of the next generation. More than ever we will have to create social mechanisms for children's upbringing so that they can learn to

face new challenges and to live in harmony with each other and with nature-environment. We cannot keep the school and the process of education isolated from the society. Teachers cannot be the only people responsible for children's education. In the truest sense, it will take the whole village to educate the child.

We are told that the ideal school system was meant to be for a holistic development of children. Most of the schools never came close to this ideal because doing well in the examinations was a priority. Now will be the time for the holistic school to come to life everywhere because being social, being creative, being caring and being human will be critical for everyone.

As jobs in manufacturing, financial services and so on are lost to the machines, in all probability activities in which human beings take care of others, where they entertain each other, where they engage in adventure and exploration, where they grow and create with their hands, will become important. Of course, we will need surgeons, engineers and researchers and they could be needed in large numbers. But, also visual arts, performing arts, athletics and sports along with foundational skills for learning and adapting will be important.

I have felt for some time that the non-linear character of the digital technology is reflecting in every sphere of our activity. It is reshaping our whole society once again. The industrial revolution once created a very organized society out of a disorganized agrarian life. Now we are changing again.

But, our education system refuses to budge from its assembly line-like, bureaucracy-influenced linear nature. What we learn, how we learn and when has to change. We do not need to discard and throw away everything we learn today. Learning to listen, read and comprehend will still be foundational. Learning to analyse solve problems will still be important. Above all, learning to learn, to be creative, to be human will have to be the most important universal outcomes of the next phase of education.

Footnotes:

1. <https://economictimes.indiatimes.com/industry/services/education/india-is-in-the-middle-of-an-engineering-education-crisis/articleshow/63680625.cms>
2. Article IV, Jomtien Declaration, 1990.
3. <https://www.bloombergquint.com/union-budget-india/2017/01/23/budget-2017-few-jobs-for-young-india-youth-unemployment-is-twice-the-national-average>
4. <https://www.cnn.com/2014/02/20/youth-unemployment-in-china-a-crisis-in-the-making.html>
5. https://www.edge.org/conversation/kai_fu_lee-we-are-here-to-create

